

Kiran George

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PROFESSIONAL SUMMARY

Generative AI Engineer with 3+ years of experience architecting and deploying production-level LLM, agentic and computer vision systems. Proven track record in developing scalable RAG pipelines, optimizing model inference speed by over 25%, and building open-source orchestration frameworks adopted by enterprise clients. Eager to leverage expertise in Gen-AI systems and cloud infrastructure to drive AI innovation.

SKILLS

- **Languages:** Python, C++, CUDA C/C++
- **AI/ML Frameworks:** PyTorch, TensorFlow, Keras, LangChain, CrewAI, Google ADK, Microsoft Autogen
- **Computer Vision:** OpenCV, Open3D, COLMAP, YOLO, Segment Anything (SAM)
- **Infrastructure & Ops:** Docker, ONNX, TensorRT, Celery, FastAPI, Flask, Git, CI/CD (GitHub Actions)
- **Cloud & Databases:** AWS (Bedrock, SageMaker, S3, EC2), PostgreSQL, Vector Databases

PROFESSIONAL EXPERIENCE

AI Engineer

Fission Labs, Hyderabad, India

Sep 2022 – Present

Engineered and deployed end-to-end Generative AI solutions for enterprise clients, leading projects from proof-of-concept to full-scale production systems on AWS. Specialized in building complex, multi-agent systems and RAG pipelines, improving information retrieval accuracy and reducing operational costs.

Key Projects:

- **Flotorch (LLM & Agentic Orchestration Framework):**
 - Co-developed and maintain Flotorch, an open-source framework for LLM orchestration, adopted in PoCs for multiple US-based clients.
 - Engineered a modular SDK for dynamic agent orchestration, enabling developers to prototype and deploy multi-agent workflows 50% faster across 4+ frameworks.
 - Architected and maintained a robust RAG orchestration system on AWS, integrating with providers including Bedrock, SageMaker, Anthropic, and OpenAI.
 - Refactored the evaluation library to enhance modularity, and developed an asynchronous agentic evaluation SDK with custom metrics, reducing evaluation-related bug reports by 30%.
- **Product Evaluation System (Walmart PoC):**
 - Designed and implemented an LLM-based product evaluation module using Claude Sonnet on AWS Bedrock, automating 80% of manual assessments.
 - Engineered a distributed task queue with Celery for parallel LLM inferences, increasing throughput by 300% vs sequential processing.
 - Developed and refined prompts, achieving a 95% acceptance rate on evaluations and securing project-wide adoption.
- **3D Measurement Solution (Valet Closet):**
 - Researched and developed a photogrammetry-based 3D object measurement solution using COLMAP and Open3D, reducing client storage costs by 40%.

- Optimized the inference pipeline with MobileSAM, ONNX, and TensorRT, improving processing speed by 25% on edge devices.
- Containerized the pipeline with Docker and exposed it via FastAPI RESTful API for seamless integration.
- Received a Spot Award for innovative research and project delivery within a 20-person team.

AI Engineer Intern

Fission Labs, Hyderabad, India

Jun 2022 – Sep 2022

Conducted foundational research into multi-view geometry and photogrammetry, producing a technical report that guided the development of the company's 3D reconstruction service.

Computer Vision Intern

Artenal Agrotech, Vancouver, Canada (Remote)

Nov 2021 – May 2022

Fine-tuned a YOLOv5 object detection model on a custom dataset of 5,000+ images, achieving 92% mAP for fruit detection in a robotic harvesting pipeline. Researched and implemented SLAM algorithms with Nvidia Isaac SDK and SIM, contributing to the autonomous navigation of the prototype robot.

PROJECTS

• **Daxch: Agent-Based Trading Assistant – Ongoing**

Architected a multi-agent stock analysis platform using Strands Agents for real-time evaluation based on technicals, sentiment, and volatility. Built a zero-cost data pipeline that ingests and caches daily market data, eliminating paid API needs. Deployed on AWS Amplify and Elastic Beanstalk with RDS and a credit-based access system.

• **NLP-Powered Air Quality Prediction System**

Developed and trained deep learning time-series models (LSTM, Conv-LSTM) on extensive air quality data, achieving 85% accuracy. Deployed a public-facing web app on Heroku for real-time air quality forecasting in Kochi, India.

• **CaptionQuest**

Built an AI system utilizing MiniCPM for automatic video/image captioning and a metadata-based content retrieval system using sentence embeddings. Experimented with multimodal models like Moondream and integrated MiniCPM, creating a pipeline for media handling, captioning, embedding, and retrieval.

EDUCATION

Master of Computer Science, Computational Perception and Robotics

Georgia Institute of Technology (GaTech), USA

2024 – Present

- *Relevant Coursework:* Human-Computer Interaction, Computational Photography, Knowledge AI

Bachelor of Engineering, Mechatronics

Nehru College of Engineering and Research Center, Kerala, India

2015 – 2019

- *Relevant Coursework:* Digital Image Processing, Robotics, Machine Vision, Embedded Systems

CERTIFICATIONS

- Accelerated Computing in CUDA C/C++ (Nvidia Deep Learning Institute, 2024)
- TensorFlow Developer Professional Certificate (deeplearning.ai, 2021)
- Machine Learning by Stanford University (Coursera, 2021)
- Programming in Modern C++ (NPTEL, 2023)